1. In Java arrays are

A. objects ✔

B. object references

C. primitive data type

D. None of the above

2. Which one of the following is a valid statement?

A. ○ `char[] c = new char();`

B. ◉ `char[] c = new char[5];` ✔

C. ○ `char[] c = new char(4);`

D. ○ `char[] c = new char[];`

3. **What is the result of compiling and running the following code?**

```
public class Test{
        public static void main(String[] args){
                int[] a = new int[0];
                System.out.print(a.length);
        }
}
```

A. ◉ 0 ✔

B. ○ Compilation error, arrays cannot be initialized to zero size. ✘

C. ○ Compilation error, it is a.length() not a.length ✘

D. ○ None of the above

4. What will be the output?

```
public class Test{
        public static void main(String[] args){
                int[] x = new int[3];
                System.out.println("x[0] is " + x
[0]);
        }
}
```

A. The program has a compile error because the size of the array wasn't specified when declaring the array.

B. The program has a runtime error because the array elements are not initialized.

C. **The program runs fine and displays x[0] is 0.** ✔

D. The program has a runtime error because the array element x[0] is not defined.

5. What is the output of the following code?

```
public class Test{
        public static void main(String args[]){
                double[] myList = {1, 5, 5, 5, 5,
1};
                double max = myList[0];
                int indexOfMax = 0;
                for(int i = 1; i < myList.length;
i++){
                        if(myList[i] > max){
                                max = myList[i];
                                indexOfMax = i;
                        }
                }
                System.out.println(indexOfMax);
        }
}
```

A. ○ 0

B. ◉ 1 ✔

C. ○ 2

6. What is output of the following code:

```
public class Test{
        public static void main(String[] args){
                int[] x = {120, 200, 016 };
                for(int i = 0; i < x.length; i++)
                        System.out.print(x[i] + "
");
        }
}
```

A. 120 200 16

B. 120 200 14 ✓

C. 120 200 016

D. 016 is a compile error. It should be written as 16.

7. Determine output:

```
public class Test{
        public static void main(String[] args){
                int[] x = {1, 2, 3, 4};
                int[] y = x;

                x = new int[2];

                for(int i = 0; i < x.length; i++)
                        System.out.print(y[i] + "
");
        }
}
```

A. 1 2 3 4

B. 0 0 0 0

C. 1 2 ✔

D. 0 0

E. None of these

8. **Analyze the following code and choose the correct answer.**

```
int[] arr = new int[5];
arr = new int[6];
```

A. The code has compile errors because the variable arr cannot be changed once it is assigned.

B. The code has runtime errors because the variable arr cannot be changed once it is assigned.

C. **The code can compile and run fine. The second line assigns a new array to arr.** ✓

D. The code has compile errors because we cannot assign a different size array to arr.

9. What will be the output?

```
public class Test{
        public static void main(String[] args){
                int[] a = new int[4];
                a[1] = 1;
                a = new int[2];
                System.out.println("a[1] is " + a
[1]);
        }
}
```

A. ○ The program has a compile error because new int[2

B. ○ The program has a runtime error because a[1

C. ◉ a[1] is 0 ✔

D. ○ a[1] is 1

11. **What would be the result of attempting to compile and run the following code?**

```
public class HelloWorld{
      public static void main(String[] args){
            double[] x = new double[]{1, 2, 3};
            System.out.println("Value is " + x
[1]);
      }
}
```

A. ○ The program has a compile error because the syntax new double[]{1, 2, 3} is wrong and it should be replaced by {1, 2, 3}.

B. ○ The program has a compile error because the syntax new double[]{1, 2, 3} is wrong and it should be replaced by new double[3]{1, 2, 3};

C. ○ The program has a compile error because the syntax new double[]{1, 2, 3} is wrong and it should be replaced by new double[]{1.0, 2.0, 3.0};

D. ○ The program compiles and runs fine and the output

12. **Which will legally declare, construct, and initialize an array?**

A. ○ int [] myList = {}; ✗

B. ○ int [] myList = (5, 8, 2);

C. ○ int myList [] [] = {4,9,7,0};

D. ◉ int myList [] = {4, 3, 7}; ✓

14. What is the value of a[1] after the following code is executed?

```
int[] a = {0, 2, 4, 1, 3};
for(int i = 0; i < a.length; i++)
a[i] = a[(a[i] + 3) % a.length];
```

A. ○ 0

B. ◉ 1 ✓

C. ○ 2

D. ○ 3

E. ○ 4